UPDATE 3 SUMMARY

- Wild event cards only happen once and is put to the discarded deck
- Cyanobacteria and Green algae costs 1 Phosphorus (P) when played/activated.
- Organism and Microbe Cards have been renamed
- All dead cards (except for microbe cards) will now go to a discarded deck.
- Added more producer and detritivore cards

GOAL

The goal of each player is to reach a certain amount of points of carbon, nitrogen, and phosphorus as fast as possible (adapted from the original card game) while managing these same points for survival.

CARD ARCHETYPES

1) **ORGANISMS** - Specifically plants, algae, detritivores (shredders, collectors, grazers, etc.), and consumers/predators. These cards are to consume or to be consumed but also the main source of nutrients. Can die to insufficient nutrients, and also has evolution/adaptation mechanics.

    a. **PRODUCERS**: Plant, Algae, Etc.
    Consumes nutrients— mainly carbon and nitrogen to produce oxygen (O_2). Has a fixed amount of lifespan / rounds. Can be eaten by detritivore cards played by any player.

    b. **DETRITIVORES**: Collectors, Grazers, Shredders
    Can eat alive and dead plant matter (regardless from which player), releasing small amounts of nutrients per round. Can not fully consume a producer card in one round, instead it may take multiple rounds depending how much detritivores are eating it.

2) **MICROBE** - Bacteria that can release all or a specific nutrient during decomposition. Allows increase in net release done by detritivores, returning more nutrients to be back for use. Can only be used on 1 dead card and can not be transferred or retracted back into the player's hand during use. Does not consume collected nutrient points (except for oxygen), but reduces net nutrient gain directly from the dead card.

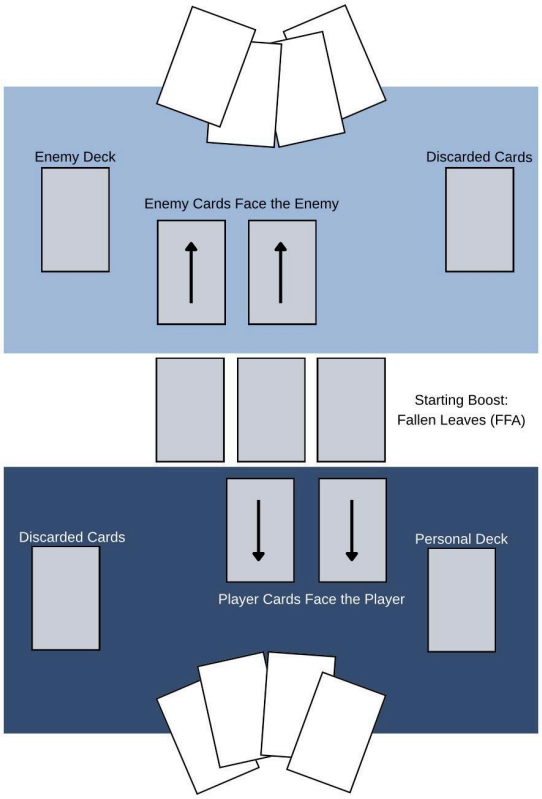
3) **WILD EVENT** - Random events that will occur to the player when drawn. Results can be advantageous or not. When drawn, the event will immediately be played but will not consume any action points. Can not stack on each other

RULES & MECHANICS

(A) Player Deck, Hand, Rounds & Turns

- There will be 4 decks in total, a separate personal deck and another separate discarded deck
- Each player will start with a hand of 5 random cards
- A player has 3 action points (fixed) per round
- A card can be disposed of for a new card, costing 1 point. Cards may also be held for future rounds
- Upon placement of any cards (except wild event cards), they consume 1 action point.
- When cards are placed/played, it can not be retracted until it finishes their action.
- Disposed cards go to a separate disposed deck, which can only be reused or reshuffled when the first deck is fully depleted.

(B) Game Set-up



(C) Card-Specific Mechanics

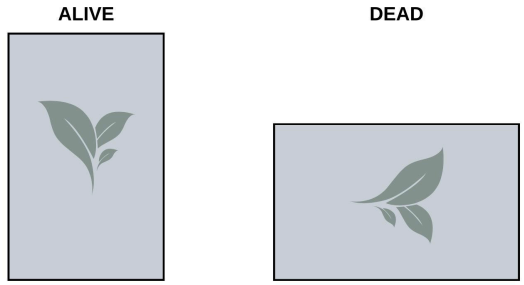
(C.1.) Organism Cards have 2 statuses: Alive & Dead

Alive - If activation conditions are satisfied, organisms will do specific processes. They may either die to the lack of necessary nutrients points

or to natural causes via a number of rounds or wild event cards. Can be consumed by other selected organisms.

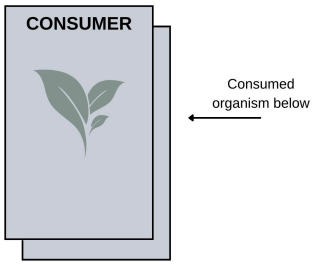
Dead / **Decaying Matter** - Organisms that can be readily be consumed by detritivore cards. Using a detritivore card will separately cost another action point in the current round but will not in the succeeding rounds. Is not immediately depleted in a single round. Complete decomposition will take multiple rounds depending on the detritivores and microbes.

(C.2.) Indication for the organism cards:



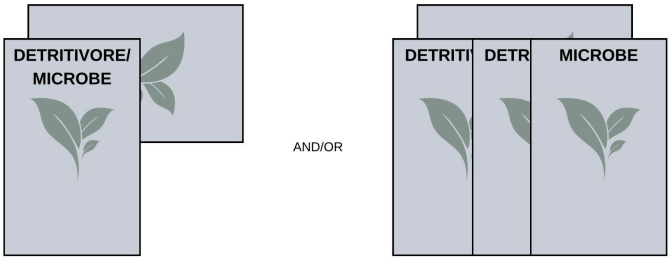
- When the nutrients of the dead cards are fully released, the card is reshuffled into the player's deck for reuse
- Detritivores stay and get extended lifespan by how many rounds it took for the dead card's biomass is fully released

(C.3.) Consumer-Specific Interactions— indications:



- A consumed organism stays below the consumer card but stays visible for tracking purposes.

(C.4.) Detritivore and Microbe-Specific Interactions— indication:



- The decaying matter getting broken down stays below any microbe or detritivore card. It must stay visible even with stacked cards for tracking purposes.

(D) Point Management System

- There are 4 kinds of nutrient points to be used and earned to the player's advantage. Three (3)— carbon, nitrogen, and phosphorus— are necessarily needed to win. One (1)— oxygen is essential for managing your cards.
- The player is given starting nutrients:
  - 10 Carbon ©
  - 3 Nitrogen (N)
  - 0 Phosphorus (P)
  - 7 Oxygen (O)
- Oxygen is not needed to win however it is used for the majority of the abilities/activation conditions.
- Points can go beyond the winning condition, but may result in consequences.

(E) Winning Conditions

- A player must achieve:
  - 50 Carbon © Points
  - 15 Nitrogen (N) Points
  - 1 Phosphorus (P) Points
- A digital sheet for tracking these points will be given beforehand

PRODUCER CARDS

Oxygen Production is per day round due to active production/photosynthesis. The ratio of carbon to oxygen at day is 1:1 except for Algae Cards

Cost refers to activation conditions, nutrients are only consumed during the first play.

Production is the net gain after a round, may contribute to player's nutrient score and/or the organisms' biomass.

Biomass is the nutrients that will be available for decomposition and nutrient cycling after dying

Plant Card	C Cost	N Cost	Lifespan	Biomass	Production to Nutrient Pool (per Round)
Duck Lettuce x2 *(Ottelia alismoides)*	-2	-0.5	3 rounds	8 C 2 N 0 P	2 O 1 C
Hydrilla x2(*Hydrilla verticillata*)	-2	-0.5	4 rounds	8 C 2 N 0 P	2 O 1 C
Water Hyacinth (*Eichhornia crassipes*)	-2	-0.5	4 rounds	8 C 2 N 0 P	2 O 1 C
Indian Lotus (*Nelumbo nucifera*)	-2	-0.5	3 rounds	8 C 2 N 0 P	2 O 1 C
Pond fern x2 *(Azolla sp.)*	-2	-0.5	3 rounds	8 C 2 N 0 P	2 O 1 C

Algal Blooms will happen if phosphorus points go past 2

Algae Card	C Cost	N Cost	P Cost	Lifespan	Biomass	Production to Nutrient Pool (per Round)
Green Algae x2 *(Chlorella sp.)*	-1	-0.5	1	2 rounds	2 C 0 N 0 P	3 O 1 C

Diatoms x3 - *Nitzschia sp.*	-2	-0.5	0	1 rounds	2 C 0 N 0 P	4 O 1 C
Cyanobacteria *Anabaena sp.*	-1	-0.5	1	2 rounds	2 C 0 N 0 P	3 O 1 C 1 N

DETRITIVORE CARDS

Shredders —> Break large dead materials
Collectors —> Consume fine organic matters
Grazers —> Consume algae/biofilm

Lifespan of detritivores can be extended for the total of rounds they are consuming dead matter. Detritivore biomass will not be added nor deducted.

Oxygen Consumption is per round. Nitrogen is a one-time use upon activation.

Released biomass goes into the player's nutrient pool.

Shredder Cards	O Cost	N Cost	Lifespan	Biomass	Ability
Common Caridina (*Caridina nilotica*)	-1	-0.5	4 rounds	3 C 1 N 0 P	**Shred:** Exclusively eats dead plant cards. Releases 3 C from a dead plant's biomass per round
Red Worm x2 (*Limnodrilus hoffmeisteri*)	-1	-0.5	3 rounds	3 C 1 N 0 P	**Burrow:** Exclusively eats dead plant cards. Releases 2 C from a dead plant's biomass per round

Collector Cards	O Cost	N Cost	Lifespan	Biomass	Ability
Sludge Worm x2	-1	-0.5	3 rounds	3 C 1 N 0 P	**Filter:** When consuming a dead plant or dead algae card, release 1 C from their biomass. When paired with a Shredder, releases an additional carbon along with a nitrogen point
Harlequin Fly x3 (*Chironomus riparius*)	-1	-0.5	3 rounds	3 C 1 N 0 P	**Fine Detritus:** When processing a dead plant or dead algae card, release 1 C from their biomass. When paired with a Shredder, releases an additional carbon along with a nitrogen point
Net-spinning	-1	-0.5	3 rounds	3 C	**Filter:** When consuming a dead

caddisfly x2(*Hydropsyche angustipennis*)				1 N 0 P	plant or dead algae card, release 1 C from their biomass. When paired with a Shredder, releases an additional carbon along with a nitrogen point

Grazer Cards	O Cost	N Cost	Lifespan	Biomass	Ability
Susong Pilipit (*Jagora asperata*)	-1	-0.5	4	3 C 1 N 0 P	**Graze:** Exclusively eats algae cards (both alive and dead). Releases 1 carbon from their biomass. Can prevent algal blooms
Mayfly Nymph Large Dark Olive (*Baetis rhodani*)	-1	-0.5	3	3 C 1 N 0 P	**Control Algae:** When consuming dead producers, remove 1 excess P associated with it. Manages phosphorus level before algal bloom becomes a problem.

MICROBE CARDS

- Return nutrients from dead organic matter to the nutrient pool.
- After assisting with decomposing, returns back to the player's hand

Card	O Cost (per round)	Biomass Consumed from Gross Released Nutrients while Recycling	Main ability
Carbon Bacteria ×3 - ***Cellulomonas fimi***	-1	-1 C -0.5 N	When recycling nutrients along with a detritivore, released carbon by detritivore is doubled but does not reduce the biomass of the dead card. Decomposes any dead organic matter. Releases 1 C per round.
Nitrifying Bacteria x3 - ***Nitrosomonas europaea***	-1	-1 C -0.5 N	When recycling nutrients along with a detritivore, it doubles and releases all nitrogen from the dead card. Decomposes any dead organic matter. Releases 1 N per round.
Phosphorus Bacteria x2 - *Bacillus megaterium*	-1	-1 C -0.5 N	When decomposing Processed Detritus, gain 1 P. Decomposes any dead organic matter. Releases 1 P per round.

Wild Event Cards

Wild Event	Effect	Rounds Affected (including current round)
Strong Rainfall	All players gain **+1 P**. If a player has more than 2 **P** and at least 1 Algae Card, trigger **Algal Bloom**. Algae Cards Biomass is doubled	1 round
Drought	During the round, Producers produce **−1 O_2**. Aquatic organisms cannot be played. Aquatic Plants cannot photosynthesize.	2 rounds
Heat Wave	All organisms consume **+1 O_2**. Organisms that cannot meet their O_2 requirement die.	2 rounds
Flash Flood	A random organism card that is already played will die. Player gains **+1 C**. If no organism is available, gain **+1 P** instead.	1 round
Introduced Non-native Species	Two random played cards (except for microbes) are instantly dead due to unwarranted competition. Nutrients are also deducted from the player: -5 C, - 2N, -0 P If there are no alive cards, the nutrient deducted is double	1 round